1. Modality Schemas & Sampling Rates

The pipeline standardises three distinct physiological modalities into a uniform [N, Channels, Time] float32 tensor structure optimized for downstream Self-Supervised Learning (SSL):

HAR (PAMAP2, WISDM, mHealth): Harmonised to a unified 6-channel IMU schema (Wrist Accelerometer X/Y/Z and Gyroscope X/Y/Z). All sets are resampled to 20 Hz, which sufficiently captures human biomechanical frequencies while aggressively reducing data redundancy.

EEG (EEGMMIDB): Filtered to a 21-channel Motor/Somatosensory subset using a standard 10-05 montage. Kept at the native 160 Hz to preserve high-frequency cortical dynamics, with a 1-45 Hz bandpass filter applied.

ECG (PTB-XL): Utilises the standard 12-lead schema. The 100 Hz signal path was explicitly chosen over the 500 Hz path. This captures the critical diagnostic features (QRS complex) while reducing the memory footprint by 80%.

2. Window Definitions, Overlaps, & Split Strategy

HAR: Split into two distinct regimes. Pretraining uses 10-second windows with 0% overlap to maximise raw data volume for contrastive learning. Supervised uses 5-second windows with 50% overlap to capture transient movements and provide dense labeled examples.

EEG: Epoched exactly to 4.0 seconds (640 samples) relative to T1/T2 motor imagery event onsets.

ECG: Maintains the standard 10-second (1000 samples) record length.

Split Strategy: ECG enforces strict patient-safe deterministic splits using the official strat_fold column (Train: Folds 1-8, Val: Fold 9, Test: Fold 10) to guarantee zero cross-split patient leakage.

3. Data Quality: Missing Values, Invalid Rows & Labels

Missing/Invalid Data: Signal dropouts or extreme artefacts in EEG (peak-to-peak amplitude > 100 µV) are algorithmically rejected using MNE drop_bad thresholds and Autoreject. However, running Autoreject algorithms is too heavy. The code for both approaches is provided, and the user can choose which one they prefer to use. Incomplete windows (e.g., file edges) are automatically trimmed or dropped.

Null & Transient Labels: In the HAR datasets, transient activities (e.g., Activity ID 0 in PAMAP2) or unmapped labels are explicitly dropped. This prevents useless classes from polluting the semantic clusters during downstream supervised fine-tuning.

4. Compute Profile & Resource Management

Peak RAM & Storage: The pipeline is highly optimised, requiring a peak RAM (Resident Set Size) of only ~2 GB.

Chunking & Streaming: To achieve this low RAM footprint, the ECG processor uses memory-efficient pre-allocation (np.empty), loading arrays directly into pre-defined memory slots rather than appending to massive lists. Similarly, HAR data processes files iteratively to prevent Out-Of-Memory crashes before final concatenation.

5. Anticipated Engineering Risks & Mitigations

Risk of File-Path Fragility (Extraction nesting): Different OS extraction tools (or wget recursive downloads) create unpredictable nested folder hierarchies (e.g., "zips inside zips"), which causes hardcoded path failures.

Mitigation: Implemented a dynamic get_actual_path() directory crawler that automatically hunts for unique dataset markers (e.g., ptbxl_database.csv), making the pipeline immune to folder-structure variations.

Risk of Target Leakage (ECG): Subjects in PTB-XL often have multiple ECG records. A naive random split would place records from the same patient into both Train and Test sets, artificially inflating model performance.

Mitigation: Engineered a strict programmatic grouping via the metadata folds, backed by an algorithmic intersection test in validation.py to mathematically prove 0% subject leakage.

Risk of Phase Distortion (EEG): Applying bandpass filters to EEG data can introduce phase shifts, disrupting temporal alignments in SSL.

Mitigation: MNE's default zero-phase (forward-backward) FIR filters were utilised to preserve true temporal relationships in the cortical data.